Sofiane Ezzehi

PhD student in Applied Mathematics

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Research interests

Model order reduction for nonlinear parametric problems; variational inequalities; hyper-reduction; semi-smooth Newton methods; scientific machine learning (neural-network-augmented reduced models, neural PDE solvers).

Education

2024 – 2027 (expected) **PhD in Applied Mathematics** — Marne-la-Vallée, France
CERMICS, École nationale des ponts et chaussées – Modelling, Analysis and Simulation (MAS) group, in collaboration with IFP Energies nouvelles (IFPEN).
Topic: Model order reduction for nonlinear parametric variational inequalities, with applications to CO_2 storage simulation.
Advisors: Virginie Ehrlacher (CERMICS) and Guillaume Enchéry (IFPEN).
Member of the MATHERIALS project-team, Inria Paris.

2023 – 2024 **MSc Mathématiques, Vision, Apprentissage (MVA)** — Paris, France
École normale supérieure Paris-Saclay. Average grade: 18.2/20.
Selected courses: computational statistics, probabilistic graphical models, reinforcement learning, graphs in machine learning, geometric data analysis.

2020 – 2024 **Engineering degree (Diplôme d'ingénieur)** — Paris, France
École nationale des ponts et chaussées – Department of Applied Mathematics and Computer Science.
Selected courses: Fourier analysis, convex optimisation, evolution problems, high-dimensional statistics, analysis and scientific computing.

Publications and preprints

2026 **S. Ezzehi**, V. Ehrlacher, G. Enchéry and T. Faney. *A hyper-reduced neural network-augmented semi-smooth Newton method for nonlinear parametric variational inequalities.* Preprint, arXiv:2607.21242.

Research experience

Oct 2024 – present **Doctoral researcher, CERMICS & IFPEN** — Marne-la-Vallée, France

- Reduced-order models for parametric variational inequalities arising in porous media with faults, motivated by CO_2 storage simulations.
- Combination of projection-based reduction (POD), neural-network-augmented reduced models and hyper-reduction within a semi-smooth Newton framework.

Apr 2024 – Jul 2024 **Research intern, CERMICS** — Marne-la-Vallée, France
Galerkin neural networks.

- Literature review of neural-network PDE solvers: PINNs, Deep Ritz, VPINNs, Galerkin neural networks.
- Implementation and comparative study on 2D Poisson problems on regular and irregular domains.

Sep 2022 – Feb 2023 **Research intern, CEA** — Paris-Saclay, France
Coupled thermo-neutronic problem.

- Development of a shooting-method algorithm extending the analysis of the coupled problem to molten salt reactors.

Talks and posters

Contributed talks

Jun 2026 CANUM 2026, 47th French National Congress on Numerical Analysis, Saint-Jacut-de-la-Mer.

Jun 2025 ADMOS 2025, XII Int. Conference on Adaptive Modeling and Simulation, Barcelona, Spain.

Dec 2025 Workshop on Industrial Applications of Numerical Analysis and Machine Learning, CWI Amsterdam.

Posters

Dec 2025 — Workshop on Industrial Applications of Numerical Analysis and Machine Learning, CWI, Amsterdam, Netherlands.

Jul 2024 — Workshop SciML on Scientific Machine Learning, Université de Strasbourg, France.

Honours and awards

2025 — **Poster Prize**, Workshop on Industrial Applications of Numerical Analysis and Machine Learning, CWI Amsterdam.

2022 — 4th of 60 teams, KIRO operations research challenge (École des ponts, CERMICS and Pelico).

2020 — **French Government Scholarship**, Ministry for Europe and Foreign Affairs – one of 8 awarded for results in the Grandes Écoles entrance exams (tuition and living expenses).

Teaching and supervision

Teaching, École nationale des ponts et chaussées (76 h in total)

Feb – Apr 2026 — **Practice of machine learning methods (PRAMA)** — 12 h
Practical sessions in Python. Random forests, gradient boosting, support vector machines and neural networks applied to real datasets.

Nov 2025 – Jan 2026 — **Analysis and PDEs, Part I (first-year course)** — 20 h
Lectures and exercise sessions. Normed vector spaces, Banach and Hilbert spaces, measure theory and Lebesgue integration.

Nov 2024 – Jan 2025 — **Analysis and PDEs, Part II (first-year course)** — 20 h
Lectures and exercise sessions. Lebesgue spaces, Fourier series, distribution theory, Fourier transform.

Supervision of first-year research projects, École nationale des ponts et chaussées

Feb – May 2026 — **Neural networks for solving partial differential equations** — 12 h
Proposed and supervised the project. Neural networks implemented in Python, then used to solve 1D/2D PDEs with physics-informed neural networks (PINNs) and the Deep Ritz method; design and validation of numerical experiments against classical methods.

Feb – May 2026 — **Model order reduction: POD and hyper-reduction (DEIM).** — 12 h
Proposed and supervised the project. Finite element solution of simple parametric PDEs (variational formulation, Lax–Milgram, Galerkin method), then construction of hyper-reduced reduced-order models with POD and D-EIM, implemented in Python.

Schools and workshops attended

Jan 2026 — XXII Jacques-Louis Lions Winter School on Numerical Simulation in Physics and Engineering, Institut Henri Poincaré, Paris.

Oct 2025 — Autumn School on Scientific Machine Learning and Numerical Methods, CWI, Amsterdam.

Jun 2024 — CEA–EDF–Inria Summer School on Model Reduction, EDF Lab, Palaiseau.

Scientific service

2026 — Member of the organizing committee, CJCMA 2026 (Congrès des Jeunes Chercheurs en Mathématiques Appliquées), École des ponts.

Skills

Scientific computing — Finite element methods (FreeFEM), reduced-basis and POD methods, hyper-reduction, semi-smooth Newton solvers, neural PDE solvers.

Languages — French (bilingual), English (fluent), Arabic (native: Tunisian; fluent: Modern Standard), Spanish (elementary).

References

Virginie Ehrlacher (PhD advisor)
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Guillaume Enchéry (PhD co-advisor)
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